

② If $f(x) = \begin{cases} xe^{-x^2/2}, & x \geq 0 \\ 0, & x < 0 \end{cases}$, then s.t. $f(x)$ is a pdf and find its cdf $F(x)$.

Soln

I.P $f(x)$ is a pdf

(i) I.P $\int_{-\infty}^{\infty} f(x) dx = 1$

(ii) F.P $\int_0^{\infty} xe^{-x^2/2} dx = 1$

LHS $\int_0^{\infty} xe^{-x^2/2} dx$

$\Rightarrow f(x)$ is a pdf

Using $F(x) = P(X \leq x) = P(-\infty < X \leq x)$

$$= \int_{-\infty}^x f(x) dx$$

Put $x^2/2 = t$

$$\frac{x dx}{x} = dt$$

$$x dx = dt$$

When $x=0$, $t=0$

$x \rightarrow \infty$, $t \rightarrow \infty$

$$\therefore \int_0^{\infty} e^{-t} \cdot dt = -e^{-t} \Big|_0^{\infty}$$

$$= -[0-1]$$

$$= 1$$

$$= \text{RHS}$$

$$= \int_{-\infty}^x x e^{-x^2/2} dx \quad \text{put } \frac{x^2}{2} = t \quad \begin{array}{l} \text{When} \\ x \rightarrow -\infty, t \rightarrow \infty \\ x \rightarrow x, t \rightarrow \frac{x^2}{2} \end{array}$$

$$\Rightarrow x dx = dt$$

$$= \int_{\infty}^{\frac{x^2}{2}} e^{-t} \cdot dt = \left[-e^{-t} \right]_{\infty}^{\frac{x^2}{2}}$$

$$= - \left[e^{-x^2/2} - 0 \right] = -e^{-x^2/2}$$

(ie) $F(x) = -e^{-x^2/2}$

↓ CDF

$$\frac{x^2}{2} \quad 0 \quad 1 \quad 2 \dots$$

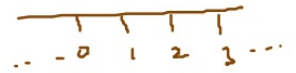
⑧ The probability distribution function of a

RV X is $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$. find the

CDF of X .

Soln

When $x < 0$, $F(x) = 0$



$$0 < x < 1, F(x) = \int_0^x f(u) du$$

$$= \int_0^x u du = \frac{u^2}{2} \Big|_0^x = \frac{x^2}{2}$$

$$1 < x < 2, F(x) = \int_0^1 u du + \int_1^x (2-u) du$$

$$= \frac{u^2}{2} \Big|_0^1 + \left[2u - \frac{u^2}{2} \right]_1^x$$

$$= \frac{1}{2} + 2x - \frac{x^2}{2} - 2 + \frac{1}{2}$$

$$= 2x - \frac{x^2}{2} - 1$$

$$x > 2, F(x) = 1$$

$$\therefore f(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{2}, & 0 < x < 1 \\ 2x - \frac{x^2}{2} - 1, & 1 < x < 2 \\ 1, & x > 2. \end{cases}$$

required
cumulative
distribution
for.

Note:
(i) Mean = $\begin{cases} \sum_x x p(x) & (\text{discrete}) \\ \int_{-\infty}^{\infty} x f(x) dx & (\text{continuous}) \end{cases}$

(ii) variance = $\begin{cases} \sum_x (x - \text{mean})^2 \cdot p(x) & (\text{discrete}) \\ \int_{-\infty}^{\infty} (x - \text{mean})^2 f(x) dx. & (\text{continuous}) \end{cases}$

① If a RV x has the pdf $f(x) = \begin{cases} \frac{x+1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$
then find the mean and variance.

Soln

$$\text{mean} = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_{-1}^1 x \cdot \left(\frac{x+1}{2}\right) dx$$

$$= \frac{1}{2} \int_{-1}^1 (x^2 + x) dx = \frac{1}{2} \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\left(\frac{1}{3} + \frac{1}{2}\right) - \left(-\frac{1}{3} + \frac{1}{2}\right) \right]$$

$$= \left(\frac{1}{3}\right)$$

$$\text{variance} = \int_{-\infty}^{\infty} (x - \text{mean})^2 \cdot f(x) dx$$

$$= \int_{-1}^1 \left(x - \frac{1}{3}\right)^3 \cdot \left(\frac{x+1}{2}\right) \cdot dx$$

$1 - \frac{2}{3} = \frac{1}{3}$

$$\downarrow$$

$$\left(x^2 + \frac{1}{9} - \frac{2}{3}x\right) \left(\frac{x+1}{2}\right)$$

$$\frac{1}{2} \left[x^3 + \frac{x}{9} - \frac{2}{3}x^2 + x + \frac{1}{9} - \frac{2}{3}x \right]$$

$\frac{1}{9} - \frac{2}{3} = -\frac{5}{9}$

$$= \frac{1}{2} \int_{-1}^1 \left(x^3 + \frac{1}{3}x^2 - \frac{5}{9}x + \frac{1}{9}\right) dx$$

$$= \frac{1}{2} \left[\int_{-1}^1 x^3 dx + \int_{-1}^1 \frac{1}{3}x^2 dx - \frac{5}{9} \int_{-1}^1 x dx + \int_{-1}^1 \frac{1}{9} dx \right]$$

$$= \frac{1}{2} \left[\frac{2}{3} \left(\frac{x^3}{3}\right)' + \frac{2}{9} [x]' \right]_0 = \left(\frac{2}{9}\right)$$

② For the density function $f(x) = ae^{-|x|}$, $-\infty < x < \infty$
find (i) value of "a" (ii) mean and variance.

Soln

$f(x) = ae^{-|x|}$ is an even function.

(i) $f(-x) = f(x)$.

$\therefore$ using $\int_{-\infty}^{\infty} f(x) dx = 1$

(i) $2 \int_0^{\infty} f(x) dx = 1$

(i) $2 \int_0^{\infty} ae^{-|x|} \cdot dx = 1$

(i) $2 \int_0^{\infty} ae^{-x} \cdot dx = 1$

mean = $\int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} x \frac{1}{2} e^{-|x|} dx$

$= \frac{1}{2} \left[\int_{-\infty}^0 x e^x dx + \int_0^{\infty} x e^{-x} dx \right]$

$$2a \left[-e^{-x} \right]_0^{\infty} = 1$$

$$-2a [0 - 1] = 1$$

$$2a = 1$$

$$\Rightarrow a = \frac{1}{2}$$

$$u = x$$

$$du = dx$$

$$du = e^x dx$$

$$u = e^x$$

$$du = e^{-x} dx$$

$$u = -e^{-x}$$

$$\therefore \frac{1}{2} \left[\int_0^{\infty} x e^x dx - \int_{-\infty}^0 x e^x dx \right] + \frac{1}{2} \left[\int_0^{\infty} -x e^{-x} dx + \int_0^{\infty} e^{-x} dx \right]$$

$$= \frac{1}{2} \left[e^x \right]_{-\infty}^0 + \frac{1}{2} \left[-e^{-x} \right]_0^{\infty} = \frac{1}{2} [1 - 0 - 0 - 1]$$

$$\text{Variance} = \int_{-\infty}^{\infty} (x - \text{mean})^2 \cdot f(x) dx = 0$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x^2 e^{-|x|} dx = \frac{1}{2} \cdot 2 \int_0^{\infty} x^2 e^{-x} dx = \underline{\underline{\int_0^{\infty} x^2 e^{-x} dx}}$$

$$u = x^2 \quad du = 2x dx$$

$$v = e^{-x} \quad dv = -e^{-x} dx$$

$$= -\cancel{x^2} e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} \cdot 2x dx$$

$$u = x \quad dv = e^{-x} dx$$

$$du = dx \quad v = -e^{-x}$$

$$= 2 \left[-\cancel{x} e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} dx \right]$$

$$= 2 \left[-e^{-x} \right]_0^{\infty} = -2 \left[0 - 1 \right] = \textcircled{2}$$

Note: 1

$$\int_0^{\infty} x^2 e^{-x} dx = 2 = 2!$$

$$\Rightarrow \int_0^{\infty} x^n e^{-x} dx = n!, \quad \forall n \geq 0$$

$\textcircled{\frac{n!}{e}}$

①

① Find the value of k and hence find mean and variance of $df = kx^2 e^{-x} dx, 0 < x < \infty$.

soln

Given $df = kx^2 e^{-x} dx$

$$\Rightarrow \frac{df}{dx} = kx^2 e^{-x}.$$

$\Downarrow$ p.d.f

(1) Given $f(x) = kx^2 e^{-x}, 0 < x < \infty$.

To find k, consider $\int_{-\infty}^{\infty} f(x) dx = 1$

$$(1) \int_0^{\infty} kx^2 e^{-x} dx = 1$$

$$\Rightarrow \boxed{k = \frac{1}{2}} \text{ (imp ①)}$$

$$\text{mean} = \int_{-\infty}^{\infty} x f(x) dx = \frac{1}{2} \int_0^{\infty} x^3 e^{-x} dx$$

$\rightarrow$ (use note ①)

$$\text{variance} = \int_{-\infty}^{\infty} (x - \text{mean})^2 f(x) dx$$

$\rightarrow$ to evaluate note ①.

Ans: mean = 3 ; variance = 3.

2) verify whether the foll. is a distribution fun.

$$F(x) = \begin{cases} 0, & x < -a \\ \frac{1}{2} \left(\frac{x}{a} + 1 \right), & -a < x < a \\ 1, & x \geq a \end{cases}$$

HINT
 $\frac{dF}{dx} = \text{pdf} \rightarrow f(x)$

$f(x) = \begin{cases} \frac{1}{2a}, & -a < x < a \\ 0, & \text{otherwise} \end{cases}$

TOP $F(x)$ is a CDF

(i) J.P $\frac{dF}{dx} = f(x)$ is a pdf

(ii) J.P $\int_{-\infty}^{\infty} f(x) dx = 1$

(iii) J.P $\int_{-a}^a \frac{1}{2a} dx = 1$

③ Find the mean and variance of the given pdf
 $f(x) = \frac{1}{4}, -2 < x < 2$.

Ans: mean = 0, variance = $\frac{4}{3}$

④ If the pdf of a RV X is $f(x) = \frac{x}{2}, 0 \leq x \leq 2$
find $P(X > 1.5 / X > 1)$.

(or) find the conditional probability of
 $X > 1.5$ given $X > 1$.

soln

$$P(x > 1.5 / x > 1) = \frac{P[(x > 1.5) \cap (x > 1)]}{P(x > 1)}$$



$$= \frac{P(1.5 < x < 2)}{P(x > 1)}$$

$$(1c) = \frac{\int_{1.5}^2 \frac{x}{2} dx}{\int_1^2 \frac{x}{2} dx} = 0.5833$$

$$= \frac{\int_{1.5}^2 f(x) dx}{\int_1^2 f(x) dx}$$

Two-dimensional random variables

So far, we have considered only one dimensional RV, in many situations, it may be interested in recording two or more characteristics of the outcome of a random experiment.

(e.g.) consider the experiment of tossing a coin twice.

$$S = \{HH, HT, TH, TT\}$$

Let X - denote the no. of heads obtained in the 1st toss.

& Y - " " " 2nd toss.

Then

S	HH	HT	TH	TT
$X(S)$:	1	1	0	0
$Y(S)$:	1	0	1	0

(X, Y) is a two dimensional RV (or) bivariate RV.
The range space $\{(1,1), (1,0), (0,1), (0,0)\}$, which is

finite $\therefore (x, y)$ is a discrete RV.

If (x, y) can assume all values in a specified region R in the xy -plane, then (x, y) is called a continuous RV.

Joint probability mass function

If (x, y) is a 2-D discrete RV $\exists$
 $P(x = x_i, y = y_j) = p_{ij}$, then p_{ij} is called the joint pmf, provided

$$(i) \quad p_{ij} \geq 0, \quad \forall i, j$$

$$(ii) \quad \sum_i \sum_j p_{ij} = 1.$$

* Since $P(x, y)$ is depending on both x and y .

But if we are interested in knowing the fu. depending only on x (or) only on y , then such a function is said to be a marginal probability function.

* The marginal probability function of x is

$$P(X=x_i) = P(X=x_i, Y=y_1) + P(X=x_i, Y=y_2) + \dots$$

$$= P_{i1} + P_{i2} + \dots$$

$$= \sum_j P_{ij} = P_{i.}$$

* The marginal probability function of y is

$$P(Y=y_j) = P(X=x_1, Y=y_j) + P(X=x_2, Y=y_j) + \dots$$

$$= P_{1j} + P_{2j} + \dots$$

$$= \sum_i P_{ij} = P_{.j}$$

Conditional probability functions of x and y

* The conditional probability function of x given $y=y_j$ is

$$P(X=x_i | Y=y_j) = \frac{P(X=x_i, Y=y_j)}{P(Y=y_j)}$$

↙ joint prob.

↘ marg. fn. of y

* The conditional probability function of y given $x = x_i$ is

$$P(Y = y_j | X = x_i) = \frac{P(X = x_i, Y = y_j)}{P(X = x_i)}$$

↙ joint prob.
↘ marg. prob. of X .

* Two Random variables X and Y are said to be independent if

$$P(X = x_i, Y = y_j) = P(X = x_i) \cdot P(Y = y_j)$$

for every i, j .

↓
marg. prob. of X

↓
marg. prob. of Y

① Consider two random variables x and $y \ni$ their joint probabilities are defined as

$$P(x=-1, y=0) = \frac{1}{3}, \quad P(x=0, y=1) = \frac{1}{3},$$

$$P(x=1, y=1) = \frac{1}{3}. \quad \text{Find}$$

- (i) marginal distribution functions of $x+y$
 (ii) conditional distribution function of x

Given $y=1$.

Soln

$x \rightarrow -1, 0, 1$

$y \rightarrow 0, 1$

$x \backslash y$	0	1	Total marg. prob. fn. of x
-1	$\frac{1}{3}$	0	$\frac{1}{3}$
0	0	$\frac{1}{3}$	$\frac{1}{3}$
1	0	$\frac{1}{3}$	$\frac{1}{3}$
Total marg. prob. fn. of y	$\frac{1}{3}$	$\frac{2}{3}$	①

* marginal distribution fun. of X is

$X=x_i$:	-1	0	1
$P(X=x_i)$:	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

$$\begin{aligned}P(X=-1) &= P(X=-1, Y=0) \\ &\quad + P(X=-1, Y=1) \\ &= \frac{1}{3} + 0 = \frac{1}{3}\end{aligned}$$

$$\begin{aligned}P(X=0) &= P(X=0, Y=0) + \\ &\quad P(X=0, Y=1) \\ &= 0 + \frac{1}{3} = \frac{1}{3}\end{aligned}$$

Marginal distribution fun. of Y is

$Y=y_j$:	0	1
$P(Y=y_j)$:	$\frac{1}{3}$	$\frac{2}{3}$

~~$\frac{1}{3}$~~
 ~~$\frac{1}{3}$~~

$$\begin{aligned}P(Y=0) &= P(X=-1, Y=0) + P(X=0, Y=0) + P(X=1, Y=0) \\ &= \frac{1}{3} + 0 + 0 = \frac{1}{3}\end{aligned}$$

$$\begin{aligned}P(Y=1) &= P(X=-1, Y=1) + P(X=0, Y=1) + P(X=1, Y=1) \\ &= 0 + \frac{1}{3} + \frac{1}{3} = \frac{2}{3}\end{aligned}$$

(ii) the conditional distribution function of x given $y=1$.

$$(ii) \quad P(x=x_i / y=1) = \frac{P(x=x_i, y=1)}{P(y=1)}$$

$$= \begin{cases} \frac{P(x=-1, y=1)}{P(y=1)} = \frac{0}{(2/3)} = 0 \\ \frac{P(x=0, y=1)}{P(y=1)} = \frac{(1/3)}{(2/3)} = 1/2 \\ \frac{P(x=1, y=1)}{P(y=1)} = \frac{(1/3)}{(2/3)} = 1/2 \end{cases}$$

$x=x_i / y=1 :$	-1	0	1
$P(x=x_i / y=1) :$	0	$\frac{1}{2}$	$\frac{1}{2}$

② The joint probability distributions of x and y is given by $P(x, y) = \frac{x+y}{21}$, $x = 1, 2, 3$; $y = 1, 2$.
Find the marginal distribution functions of x and y . Also find the conditional distribution of y given $x=3$.

Soln

$x \backslash y$	1	2	Total
1	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{5}{21}$
2	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{7}{21}$
3	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{9}{21}$
Total	$\frac{9}{21}$	$\frac{12}{21}$	①

The marginal distribution function of x is

$$x = x_i : 1 \quad 2 \quad 3$$

$$P(x = x_i) : \frac{5}{21} \quad \frac{7}{21} \quad \frac{9}{21}$$

The marginal distribution function of y is

$$y = y_j : 1 \quad 2$$

$$P(y = y_j) : \frac{9}{21} \quad \frac{12}{21}$$

Conditional distribution of Y given $X=3$ is

$$P(Y=y_j / X=3) = \frac{P(X=3, Y=y_j)}{P(X=3)}$$

$$(P_k) = \begin{cases} \frac{P(X=3, Y=1)}{P(X=3)} = \frac{\binom{4}{1,21}}{\binom{9}{21}} = \frac{4}{9} \\ \frac{P(X=3, Y=2)}{P(X=3)} = \frac{\binom{5}{21}}{\binom{9}{21}} = \frac{5}{9} \end{cases}$$

$\therefore$

$Y=y_j / X=3 :$	1	2
$P(Y=y_j / X=3) :$	$\frac{4}{9}$	$\frac{5}{9}$

→ conditional probability distribution of Y given $X=3$.

- ③ The joint probability distributions of x and y is $P(x, y) = k(2x + y)$, $x = 0, 1, 2$; $y = 0, 1, 2$.
Find (i) the value of k
(ii) marginal distributions of x and y
(iii) Are x and y independent variables?
(iv) the conditional distribution of y for $x = x$

Soln

$x \backslash y$	0	1	2
0	0	k	$2k$
1	$2k$	$3k$	$4k$
2	$4k$	$5k$	$6k$

For joint probability distribution,

$$\sum_x \sum_y P(x, y) = 1$$

$$(i.e.) \quad 0 + k + 2k + 2k + 3k + 4k + 4k + 5k + 6k = 1$$

$$\Rightarrow k = \underline{\underline{\frac{1}{27}}}$$

$x \backslash y$	0	1	2	Total (marginal p-f of x)
0	0	$\frac{1}{27}$	$\frac{2}{27}$	$\frac{3}{27}$
1	$\frac{2}{27}$	$\frac{3}{27}$	$\frac{4}{27}$	$\frac{9}{27}$
2	$\frac{4}{27}$	$\frac{5}{27}$	$\frac{6}{27}$	$\frac{15}{27}$
Total marginal distn. fn. of y.	$\frac{6}{27}$	$\frac{9}{27}$	$\frac{12}{27}$	①

* Marginal distn. fn. of x

$x = x_i :$	0	1	2
$P(x = x_i) :$	$\frac{3}{27}$	$\frac{9}{27}$	$\frac{15}{27}$

* Marginal distn. fn. of y

$y = y_j :$	0	1	2
$P(y = y_j) :$	$\frac{6}{27}$	$\frac{9}{27}$	$\frac{12}{27}$

T.P X and Y are independent

✓ (2) Def $P(X=x_i, Y=y_j) = P(X=x_i) \cdot P(Y=y_j)$

When $x=0, y=1$

$$\text{LHS } P(X=0, Y=1) = \frac{1}{27}$$

$$\text{RHS } P(X=0) = 3/27 \Rightarrow P(X=0) \cdot P(Y=1)$$

$$P(Y=1) = 9/27 = \frac{1}{3}$$

$\therefore$ ① is true for $x=0, y=1$.

But when $x=0, y=0$:

LHS $p(x=0, y \geq 0) = 0$

RHS $P(X=0) = 3/27$ $\neq P(Y=0) = 6/27$

$$\Rightarrow \underset{\text{RHS}}{P(X=0) \cdot P(Y=0)} \neq 0 = \text{LHS}$$

$\therefore$ when $x=0, y=0$, ① is not true-

$\Rightarrow X \text{ \& } Y$ are not independent

* Conditional distribution of y for $x = x_i$.

$$P(Y = y_j / X = x_i) = \frac{P(X = x_i, Y = y_j)}{P(X = x_i)}$$

when $x=0$, $\frac{P(X=0, Y=y_j)}{P(X=0)}$

$$P(X=0) \rightarrow \frac{3}{27}$$

$$\frac{\frac{1}{27}}{\frac{3}{27}} = \frac{1}{3}$$

* when $x=1$,

$$\frac{P(X=1, Y=y_j)}{P(X=1)}$$

$$P(X=1)$$

$$\rightarrow \frac{9}{27}$$

* when $x=2$,

$$\frac{P(X=2, Y=y_j)}{P(X=2)}$$

$$P(X=2)$$

$$\rightarrow \frac{15}{27}$$

$x \backslash y$	0	1	2
$P(Y/X=0)$	0	$\frac{1}{3}$	$\frac{2}{3}$
$P(Y/X=1)$	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{4}{9}$
$P(Y/X=2)$	$\frac{4}{15}$	$\frac{5}{15}$	$\frac{6}{15}$

① The joint probability function of x and y is $p(x, y) = \frac{x+y}{21}$, $x=1, 2, 3$; $y=1, 2$. Find the means of x , y , xy and $x+y$.

sols

$$\text{mean of } x = \sum x p(x) \quad \leftarrow \text{marginal}$$

$$\text{mean of } y = \sum y p(y) \quad \leftarrow \text{marginal}$$

$$\text{mean of } xy = \sum \sum (xy) p(x, y) \quad \leftarrow \text{joint pmf.}$$

$$\text{mean of } (x+y) = \sum \sum (x+y) p(x, y) \quad \leftarrow \text{joint pmf.}$$

$x \backslash y$	1	2	total	
1	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{5}{21}$	* marginal distn for of x is
2	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{7}{21}$	
3	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{9}{21}$	
total	$\frac{9}{21}$	$\frac{12}{21}$		$x = x_i: \quad 1 \quad 2 \quad 3$ $P(x = x_i): \quad \frac{5}{21} \quad \frac{7}{21} \quad \frac{9}{21}$

* marginal distn. for
of y is

$y = y_j: \quad 1 \quad 2$
 $P(y = y_j): \quad \frac{9}{21} \quad \frac{12}{21}$

mean of x

$$= \sum x P(x)$$

$$= \frac{5}{21} + \frac{14}{21} + \frac{27}{21} = \frac{46}{21}$$

mean of $y = \sum y P(y)$

$$= \frac{9}{21} + \frac{24}{21} = \frac{11}{7}$$

mean of $xy = \sum \sum (xy) P(x, y)$

$$= (1)(1)\left(\frac{2}{21}\right) + (1)(2)\left(\frac{3}{21}\right) + \dots + (3)(2)\left(\frac{5}{21}\right)$$

$$= \frac{24}{7}$$

* mean of $(x+y) = \sum \sum (x+y) P(x,y)$

$$(19) = (2) \frac{2}{21} + (3) \frac{3}{21} + (3) \frac{3}{21} + (4) \frac{4}{21} \\ + (4) \left(\frac{4}{21} \right) + (5) \left(\frac{5}{21} \right)$$

$$= \frac{79}{21}$$

② The Bivariate Probability function of (x,y) is given below. find $P(x \leq 1)$, $P(y \leq 3)$, $P(x+y \leq 4)$.

Soln

(i) $P(x \leq 1)$

$$= P(x=0) + P(x=1)$$

$$= \frac{1}{4} + \frac{5}{8} = \frac{7}{8}$$

$x \backslash y$	1	2	3	4	5	6	Total
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{1}{4}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{5}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{1}{8}$
Total	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	①

(ii) $P(y \leq 3)$

$$= P(y=1) + P(y=2) + P(y=3)$$

$$= \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

(iii) $P(X+Y \leq 4)$.

$X+Y \leq 4$ is satisfied by

$$\left. \begin{array}{l} (0,1), (0,2), (0,3), (0,4) \\ (1,1), (1,2), (1,3) \\ (2,1), (2,2) \end{array} \right\} P(X+Y \leq 4)$$

$$= 0 + 0 + \frac{1}{32} + \frac{2}{32}$$

$$+ \frac{1}{16} + \frac{1}{16} + \frac{1}{8}$$

$$+ \frac{1}{32} + \frac{1}{32}$$

$$= \frac{13}{32}$$

- ③ Three balls are drawn at random from a box containing 2 white, 3 red and 4 black balls. If X denotes the no. of white balls drawn & Y denotes the no. of red balls drawn. Find the joint prob. distrib. of X & Y .

Soln

Given, x - no. of white balls drawn

y - " Red "

Since 3 balls are drawn
at random,

x may be 0, 1, 2

y may be 0, 1, 2, 3

2W 3R 4B.

9 balls.



3 balls

when $x=0, y=0$,

$$\begin{aligned}\rightarrow P(x=0, y=0) &= P(0 \text{ white}, 0 \text{ red}) \\ &= P(3 \text{ black balls}) \\ &= \frac{{}^4C_3}{{}^9C_3} = \left(\frac{1}{21}\right)\end{aligned}$$

$$\begin{aligned}\rightarrow P(x=0, y=1) &= P(0 \text{ white}, 1 \text{ red}) \\ &= P(1 \text{ red}, 2 \text{ black}) \\ &= \frac{{}^3C_1 \times {}^4C_2}{{}^9C_3} = \left(\frac{3}{14}\right)\end{aligned}$$

$$\rightarrow P(x=1, y=3) = P(1W, 3R)$$

4 balls.

But we need only
3 balls

$\therefore = 0$

x/y

0

1

2

3

$\frac{1}{21}$

$\frac{3}{14}$

0

.

0

1

0

0

2

